

PROJECT BASED LEARNING REPORT

NATURAL LANGUAGE PROCESSING



Title:
Detecting Hoax News in Bahasa Indonesia using IndoBERT

By:

Class: LC02

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CHAPTER 1

INTRODUCTION

1.1 Background

Hoaxes and misinformation continue to spread rapidly through social media and online news portals in Indonesia. Automatic detection for Bahasa Indonesia helps fact-checkers, journalists, and the general public to filter and verify content faster. This project builds a modern, explainable NLP pipeline focusing on Indonesian-language fake news detection with practical constraints: limited labeled data, need for interpretability, and deployment feasibility on modest hardware.

1.2 Problem Formulation

Hoax detection can be formulated as a binary text classification problem, where news content is classified as either hoax or non-hoax. This task is challenging because of imbalanced data, variations in language use, and the difficulty of understanding context using traditional machine learning approaches. Therefore, more advanced models that can capture semantic and contextual information are required to improve classification accuracy, particularly for Indonesian-language news.

1.3 Proposed Solution

To bridge this gap, this project proposes the implementation of a semantic-aware hoax detection model using Transformer-based architectures, specifically IndoBERT and Multilingual BERT (mBERT). Unlike traditional text representation methods such as bag-of-words or TF-IDF, these models leverage contextual embeddings to capture the semantic meaning of entire sentences rather than relying solely on word frequency. By fine-tuning pre-trained BERT models on Indonesian news data, the system is able to understand contextual relationships and linguistic nuances within the text. Furthermore, a comparative evaluation between IndoBERT and mBERT is conducted to analyze the effectiveness of language-specific and multilingual models in detecting hoax content, with the goal of improving classification accuracy through deeper semantic understanding of the text.

CHAPTER 2

RELATED WORKS

Fake news and hoax detection has been popular in the natural language processing area lately, especially with the rise of social media and online news platforms. In the early studies, traditional machine learning algorithms are applied, like Naive Bayes, SVM, and Logistic Regression using TF-IDF and bag-of-words. These few methods are quite accurate in capturing contextual meaning and semantic relationship within texts.

With the improvement of deep learning, models such as CNNs and RNNs improved fake news classifications by better modeling textual sequences. However, these models still faced challenges in understanding long-range dependencies and complex linguistic patterns. Transformer-based models like BERT addresses the limitations by leveraging self-attention mechanisms to capture global contextual information more effectively.

From several studies, it shows that BERT-based models significantly outperform traditional and earlier deep learning approaches in fake news detection tasks. From researches, language-specific pretrained models often yield better results than multilingual models. For Indonesian language, IndoBERT has successfully applied various NLP tasks like, sentiment analysis, hate speech detection, etc. Comparative studies between IndoBERT and multilingual BERT consistently report superior performance from IndoBERT because of its monolingual pretraining on large Indonesian corpora.

Our project builds upon prior work by applying Transformer-based architectures to Indonesian hoax news detection and providing a direct performance comparison between IndoBERT and mBERT which results in confirming existing findings that language-specific Transformer models are more effective for Indonesian text classification tasks.

CHAPTER 3

DATASET USED

3.1 Dataset Selection

This study uses the **Indonesia False News (Hoax) Dataset** obtained from Kaggle, which contains Indonesian news articles labeled as hoax or non-hoax. The dataset is divided into two files: “**Data_latih.csv**” for training and “**Data_uji.csv**” for testing, where the training data includes both news text and labels, and the testing data contains only the news text. All labels are converted into a binary format, with **0** representing non-hoax and **1** representing hoax. Because the dataset is initially imbalanced, an upsampling technique is

applied to balance the classes. This dataset is chosen because it represents real Indonesian news content and is suitable for training and evaluating BERT-based models for hoax detection.

3.2 Final Dataset

Consequently, we selected the Indonesia False News (Hoax) Dataset from Kaggle as the final dataset for our evaluation. This choice was motivated by the need for a reliable and representative benchmark for Indonesian hoax detection:

1. **Real-World News Content:** The dataset consists of Indonesian news narratives collected from online platforms, reflecting real cases of misinformation encountered in digital media.
2. **Semantic Complexity:** Hoax news often embeds misleading information within narrative descriptions rather than explicit false statements, making this dataset suitable for evaluating semantic-aware models such as BERT.
3. **Binary Labeled Data:** Each news article is clearly labeled as hoax or non-hoax, enabling supervised learning and objective performance evaluation.
4. **Balanced Training Setup:** After applying data balancing techniques, the dataset provides an even distribution of hoax and non-hoax samples, ensuring that evaluation metrics are not biased by class imbalance.
5. **Language-Specific Evaluation:** As the dataset is written in Indonesian, it allows for an effective assessment of language-specific models such as IndoBERT as well as multilingual models like mBERT in a realistic application setting.

CHAPTER 4 PREPROCESSING

4.1 Text Cleaning and Normalization

To prepare the Indonesia False News (Hoax) Dataset for effective model training, a structured and systematic text cleaning and normalization pipeline was implemented to reduce noise and ensure full compatibility with Transformer-based language models such as IndoBERT and mBERT. This preprocessing stage is critical, as the quality of textual input directly influences the model's ability to learn meaningful semantic patterns related to hoax and non-hoax news.

The preprocessing process begins by converting all news text to lowercase. This normalization step ensures textual consistency and prevents the model from interpreting the same word differently due to variations in capitalization. By reducing vocabulary duplication, the model can focus on learning semantic relationships rather than superficial textual differences.

Next, URLs and web-related strings commonly present in online news articles are removed using regular expressions. These elements typically contain links, metadata, or promotional content that do not contribute meaningful semantic information for hoax detection. Their removal helps prevent unnecessary noise from being introduced into the training data, thereby improving model robustness. Following this step, punctuation symbols and non-alphabetic characters are removed to further reduce non-linguistic noise. This simplification preserves the core meaning of the text while making it easier for the model to process.

After cleaning the text structure, Indonesian stopwords are eliminated using the Sastrawi StopWord Remover. Stopwords such as conjunctions, prepositions, and commonly used function words appear frequently across documents but provide limited discriminative value for classification tasks. Removing these words reduces redundancy in the dataset and allows the model to focus on more informative and content-bearing terms that are more relevant for distinguishing hoax from non-hoax news.

Finally, extra whitespace is removed to ensure clean and consistent text formatting. The resulting normalized text is then stored as the final input for the tokenization stage. Tokenization itself is handled internally by the BERT tokenizer, which performs subword segmentation and generates contextualized input representations suitable for Transformer models.

Overall, this preprocessing pipeline produces a clean, normalized, and noise-reduced Indonesian news corpus. By standardizing the text while preserving its semantic content, the pipeline supports more effective contextual understanding and improves the overall performance of Transformer-based models in hoax detection tasks.

4.2 Tokenization and Input Representation

After the text cleaning and normalization stage, the processed Indonesian news narratives are converted into numerical representations that can be processed by deep learning models through BERT-based tokenization. In this study, two pretrained tokenizers are employed, namely IndoBERT-base-p1 and BERT-base-multilingual-cased (mBERT). These tokenizers are specifically designed to work with their corresponding Transformer architectures and are capable of handling subword-level representations, which is particularly important for capturing morphological variations in the Indonesian language.

During tokenization, each cleaned text input is transformed into a set of token IDs, attention masks, and segment (token type) embeddings required by the BERT model. Token IDs represent the numerical encoding of subword tokens, while attention masks indicate which tokens should be attended to by the model and which correspond to padding. Segment embeddings are included to maintain compatibility with the BERT input format, even though the task involves single-sequence classification.

To ensure consistent input dimensions, truncation and padding are applied with a fixed maximum sequence length of 128 tokens. Padding is used for shorter news articles to match the required length, while truncation is applied to longer texts to ensure they remain within the model's input constraints. This uniform sequence length allows efficient batch processing and stable training across all samples.

The tokenized data is then organized into a custom PyTorch Dataset class, which facilitates structured data handling and seamless integration with the PyTorch DataLoader. Each dataset entry consists of input token IDs, attention masks, and the corresponding class label for supervised learning. The DataLoader enables mini-batch training, shuffling, and optimized memory usage during model training and evaluation.

These tokenized representations are directly passed into the IndoBERT and mBERT models, where contextual embeddings are generated internally during the fine-tuning process. By leveraging the pretrained linguistic and semantic knowledge embedded within these Transformer models, the approach enables effective learning of contextual relationships, syntactic patterns, and semantic nuances in Indonesian news text. This eliminates the need for manual feature engineering and supports more accurate and robust hoax classification performance.

CHAPTER 5

RESULTS AND ANALYSIS

5.1 Experimental Setup

To evaluate the effectiveness of Transformer-based models for Indonesian hoax detection, we conducted experiments using two pre-trained architectures: IndoBERT (indobert-base-p1) and mBERT (bert-base-multilingual-cased). The experiments were implemented using PyTorch and the Hugging Face Transformers library.

The training configuration was standardized across both models to ensure a fair comparison:

- **Hyperparameters:**
 - **Epochs:** 2
 - **Batch Size:** 16
 - **Learning Rate:** 2e-5
 - **Optimizer:** AdamW
 - **Max Sequence Length:** 128 tokens
- **Data Handling:**
 - **Preprocessing:** Text was lowercased, and URLs, punctuation, and stopwords (using Sastrawi) were removed to reduce noise.
 - **Balancing:** We employed an upsampling technique to balance the dataset, resulting in a total of 3,465 samples for the Hoax class and 3,465 samples for the Non-Hoax class.
- **Environment:** The models were trained using GPU acceleration (CUDA) where available to expedite the fine-tuning process.

5.2 Evaluation Metrics

The performance of the models was evaluated using the following standard classification metrics derived from the confusion matrix:

1. **Accuracy:** The overall percentage of correctly classified news articles.
2. **Precision:** The proportion of predicted hoax news that was actually hoax.
3. **Recall:** The ability of the model to identify all actual hoax news in the dataset.
4. **F1-Score:** The harmonic mean of precision and recall, providing a balanced view of model performance.

5.3 Performance of IndoBERT

The IndoBERT model, which was pre-trained specifically on a massive corpus of Indonesian text, demonstrated exceptional performance.

Classification Report:

	precision	recall	f1-score	support
Non-Hoax	0.98	1.00	0.99	3465
Hoax	1.00	0.98	0.99	3465
Accuracy	-	-	0.99	6930
Macro avg	0.99	0.99	0.99	6930

Weighted avg	0.99	0.99	0.99	6930
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Analysis: IndoBERT achieved a near-perfect accuracy of **99%**.

- **Precision:** The model achieved a 1.00 (100%) precision score for the Hoax class. This indicates that every single news item flagged as a "Hoax" by the model was indeed a hoax; there were virtually zero false positives.
- **Recall:** The recall for the Hoax class was **0.98**, meaning the model successfully identified 98% of all hoax news in the dataset, missing only a negligible fraction.
- **Conclusion:** This performance confirms that IndoBERT's language-specific pre-training allows it to capture the subtle semantic patterns of Indonesian misinformation with extremely high reliability.

5.4 Performace of mBERT

To provide a baseline, we evaluated the Multilingual BERT (mBERT) model, which shares its vocabulary across 104 languages.

Classification Report:

	precision	recall	f1-score	support
Non-Hoax	0.88	0.96	0.92	3465
Hoax	0.95	0.87	0.91	3465
Accuracy	-	-	0.92	6930
Macro avg	0.92	0.92	0.92	6930

Weighted avg	0.92	0.92	0.92	6930
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While mBERT performed well with an accuracy of **92%**, it notably lagged behind IndoBERT.

- **Recall Drop:** The most significant difference was in the Recall for the Hoax class (0.87 for mBERT vs 0.98 for IndoBERT). This means mBERT failed to detect approximately 13% of the hoaxes, misclassifying them as legitimate news.
- **Language Limitation:** This suggests that while mBERT understands general sentence structures, it likely misses specific Indonesian colloquialisms or "hoax-style" phrasing (morphological nuances) that IndoBERT captures effectively.

5.5 Comparative Analysis

Discussion:

1. **Superiority of Monolingual Models:** IndoBERT outperformed mBERT by a margin of **7%** in overall accuracy. This strongly supports the hypothesis that for specialized tasks in "low-to-medium resource" languages like Indonesian, a dedicated monolingual model is far superior to a generalized multilingual model.
2. **False Negatives (Safety):** In the context of hoax detection, "False Negatives" (labeling a hoax as safe) are dangerous. IndoBERT had very few false negatives (Recall 0.98), whereas mBERT missed a significant portion (Recall 0.87). Therefore, IndoBERT is much safer for real-world deployment.
3. **Effectiveness of Balancing:** Both models achieved high macro-average scores relative to their accuracy, confirming that the upsampling strategy effectively prevented class bias.

CHAPTER 6

CONCLUSION & FUTURE WORK

In this project, we discussed about the effectiveness of Transformer-based models for detecting hoax news in Bahasa Indonesia by comparing a language-specific model (IndoBERT) with a multilingual model (mBERT). Using the Indonesia False News (Hoax) Dataset, a complete NLP pipeline was designed, including text preprocessing, tokenization, data balancing, and fine-tuning of pretrained BERT models.

From the testing experiment, it clearly shows that IndoBERT significantly outperformed mBERT across all evaluation metrics. IndoBERT achieved an accuracy of 99%, with very high precision and recall for both hoax and non-hoax classes. This indicates that language-specific pretraining allows IndoBERT to better capture Indonesian linguistic structures, contextual nuances, and hoax-related phrasing. In contrast, while mBERT performed reasonably well, its lower recall for the hoax class demonstrates limitations in detecting misinformation when relying on a shared multilingual representation.

Overall, the findings confirm that monolingual Transformer models are more suitable for specialized NLP tasks in Indonesian, particularly for safety-critical applications such as hoax detection where false negatives can have serious consequences. The applied data balancing strategy also proved effective in reducing class bias and ensuring fair model evaluation. This study demonstrates that IndoBERT is a strong and practical solution for Indonesian hoax news detection and provides a solid foundation for future research and real-world deployment in misinformation filtering systems.

Future studies may enhance this hoax detection system by incorporating a broader and more recent collection of Indonesian textual data, including social media posts, news headlines, and informal language, to better reflect real-world usage. In addition, evaluating the model across different domains, addressing challenges such as sarcasm and implicit misinformation, and deploying the system in real-time web or mobile platforms would help assess its scalability, effectiveness, and practical applicability in real-world misinformation detection scenarios.

REFERENCES

- [1] Kaggle, “Indonesia False News (Hoax) Dataset,” Kaggle Datasets. [Online]. Available: <https://www.kaggle.com>
- [2] B. Wilie *et al.*, “IndoBERT: A Pretrained Language Model for Indonesian,” *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 7574–7585.
- [3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186.
- [4] Hugging Face, “Transformers: State-of-the-Art Natural Language Processing,” 2023. [Online]. Available: <https://huggingface.co/transformers>
- [5] P. Ramadhan and A. F. Hidayatullah, “Sastrawi: Indonesian Stopword Remover and Stemmer,” GitHub Repository. [Online]. Available: <https://github.com/sastrawi/sastrawi>
- [6] A. Paszke *et al.*, “PyTorch: An Imperative Style, High-Performance Deep Learning Library,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [7] D. P. Kingma and J. Ba, “Adam: A Method for Stochastic Optimization,” *International Conference on Learning Representations (ICLR)*, 2015.
- [8] W. Widodo and M. Nugraheni, “Indonesian Fake News Classification Using Transfer Learning and IndoBERT,” *International Journal on Informatics Visualization*, vol. 8, no. 3, 2024.
- [9] A. Awalina, J. Fawaid, R. Y. Krisnabayu, N. Yudistira, “Indonesia’s Fake News Detection using Transformer Network,” arXiv:2107.06796, 2021.

Appendix A

Team Contribution Statement

1. Darren Nicholas Suwito : Proposal, Coding, PPT, Final Report
2. Kent Putra Lamiran : Proposal, PPT, Final Report, Coding Idea
3. Nathanael Reinhart: Proposal, PPT, Final Report, Coding

Appendix B

Screenshots and Code Snippet

Figure 1. Load Data and Fix Label for IndoBERT

```
import pandas as pd
import numpy as np
import re, string, torch, time
from torch.utils.data import Dataset, DataLoader
from transformers import BertTokenizer, BertForSequenceClassification, get_linear_schedule_with_warmup
from torch.optim import AdamW
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, classification_report
from Sastrawi.StopWordRemover.StopWordRemoverFactory import StopWordRemoverFactory
from tqdm.auto import tqdm
import matplotlib.pyplot as plt
import seaborn as sns

# =====
# 1. LOAD DATA
# =====
train_df = pd.read_csv("Data_latih.csv")
test_df = pd.read_csv("Data_uji.csv")

print("Kolom Data Latih:", list(train_df.columns))

# Use narasi + label
train_df = train_df[["narasi", "label"]].rename(columns={"narasi": "text"})
test_df = test_df[["narasi"]].rename(columns={"narasi": "text"})

# =====
# 2. FIX LABEL (0 = Non-Hoax, 1 = Hoax)
# =====
def fix_label(x):
    x = str(x).lower().strip()
    if x in ["1", "hoax"]:
        return 1
```

Figure 2. Balance and Clean Data

```
# =====
# 3. BALANCE DATA (GPU MODE -- FULL DATA)
# =====
hoax = train_df[train_df["label"] == 1]
nonhoax = train_df[train_df["label"] == 0]

# Balance the data so both have the same amount (Upsampling)
if len(hoax) > len(nonhoax):
    nonhoax = nonhoax.sample(len(hoax), replace=True, random_state=42)
else:
    hoax = hoax.sample(len(nonhoax), replace=True, random_state=42)

train_df = pd.concat([hoax, nonhoax]).sample(frac=1, random_state=42).reset_index(drop=True)

print("✅ USING FULL DATASET (GPU POWERED):")
print(train_df["label"].value_counts())

# =====
# 4. CLEAN TEXT
# =====
factory = StopWordRemoverFactory()
stopwords = factory.create_stop_word_remover()

def clean_text(text):
    text = str(text).lower()
    text = re.sub(r"http[s]?|www[.]+|https[.]+", "", text)
    text = text.translate(str.maketrans('', '', string.punctuation))
    text = stopwords.remove(text)
    return text.strip()

train_df["clean_text"] = train_df["text"].apply(clean_text)
```

Figure 3. Classification Report and Confusion Matrix for IndoBERT

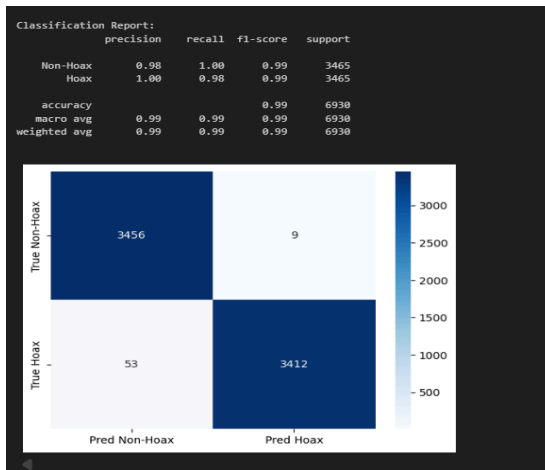


Figure 4. Training Model BERT

```
# =====
# TRAINING MODEL BERT-base-multilingual-cased (mBERT)
# =====
from transformers import BertTokenizer, BertForSequenceClassification

MBERT_NAME = "bert-base-multilingual-cased"

tokenizer_mbert = BertTokenizer.from_pretrained(MBERT_NAME)

mbert_dataset = NewsDataset(
    train_df['clean_text'].tolist(),
    train_df['label'].tolist(),
    tokenizer_mbert
)

mbert_loader = DataLoader(mbert_dataset, batch_size=16, shuffle=True)

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
mbert_model = BertForSequenceClassification.from_pretrained(MBERT_NAME, num_labels=2)
mbert_model.to(device)

optimizer = AdamW(mbert_model.parameters(), lr=2e-5)
epochs = 2
total_steps = len(mbert_loader) * epochs
scheduler = get_linear_schedule_with_warmup(optimizer, 0, total_steps)
scaler = torch.cuda.amp.GradScaler()

# ===== TRAIN =====
mbert_model.train()
for epoch in range(epochs):
    total_loss = 0
    progress_bar = tqdm(mbert_loader, desc=f"mBERT Epoch {epoch+1}/{epochs}")

    for batch in progress_bar:
        optimizer.zero_grad()
        inputs = {k: v.to(device) for k, v in batch.items()}

        with torch.cuda.amp.autocast():
            outputs = mbert_model(**inputs)
            loss = outputs.loss

        scaler.scale(loss).backward()
        scaler.step(optimizer)
        scaler.update()
        scheduler.step()

        total_loss += loss.item()
        progress_bar.set_postfix({"loss": f"{loss.item():.4f}"})

    print(f"Avg Loss mBERT: {total_loss/len(mbert_loader):.4f}")
```

Figure 5. Classification Report and Confusion Matrix for mBERT Model

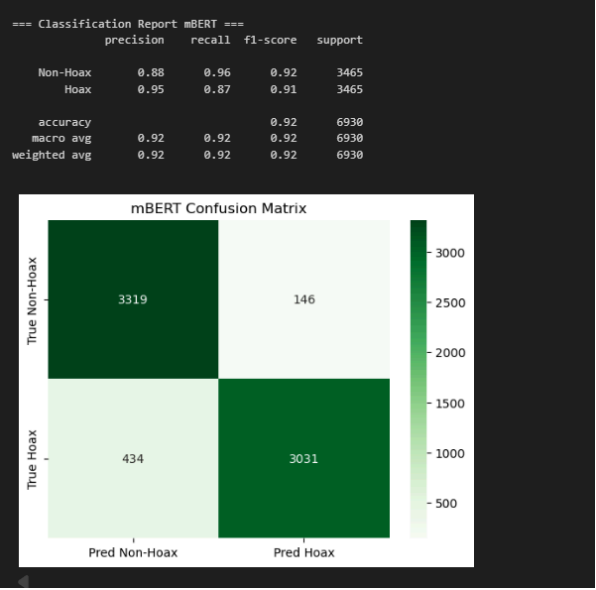


Figure 6. Prediction Result 1

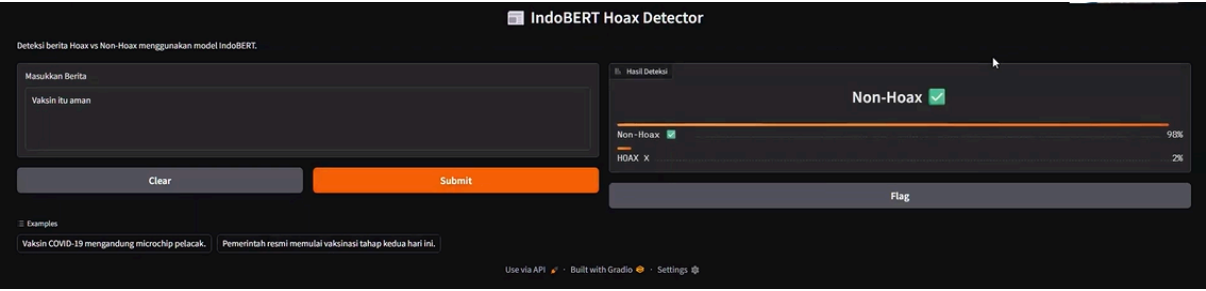


Figure 7. Prediction Result 2

